

Formula 1 Race Performance Analysis and Result Prediction

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Abstract. This study aims to analyze Formula 1 race performances and predict race results. The study covers historical Formula 1 race data from 2018–2024. Various features such as driver performance, qualifying times, in-season form, race week preparation indicators, and past race results were included in the data.

Preprocessing and feature engineering steps were applied to the obtained data, and then prediction models were created using Linear Regression and Random Forest algorithms. The performance of the models was evaluated with various regressions, and the results obtained were compared.

The developed system is integrated using a Flask-based API and a user-friendly web interface. Users can obtain predicted race finishing positions by entering parameters such as starting order, driver form, weekend performance indicators, and season momentum. The system is also supported by visualizations and performance reports that support the prediction results.

Our findings demonstrate that machine learning methods can be used as an effective tool in predicting Formula 1 race results.

Keywords: Formula 1, Race Performance Analysis, Race Prediction, Machine Learning, Natural Regression, Random Forest, Flask API, Feature Engineering.

1 Introduction

Formula 1 is arguably the most data-driven sport in the world, where microscopic margins separate victory from defeat. A driver's race outcome is never determined by a single isolated variable; instead, it is the cumulative result of a highly complex, multi-layered ecosystem encompassing qualifying execution, mechanical car reliability, constructor engineering excellence, pit stop strategies, and seasonal momentum. Since almost every facet of a Grand Prix weekend is

continuously tracked, quantified, and recorded, Formula 1 data provides an exceptionally rich yet challenging foundation for predictive machine learning workflows.

The primary objective of this project is to construct a comprehensive, end-to-end data science and software integration pipeline designed to accurately predict a driver's final race finishing position. Moving beyond isolated statistical analysis, this study implements a rigorous methodology spanning historical data from the 2018 to 2024 seasons. By utilizing the 2018–2023 seasons for model training and explicitly preserving the 2024 season as a completely held-out test set, the system evaluates how effectively predictive algorithms can generalize to unseen future race contexts under evolving regulations.

To achieve this objective, the project successfully bridges the gap between pure data science and functional software engineering through a structured, multi-disciplinary technical workflow. The architecture of this project encompasses:

- **Advanced Preprocessing & Feature Engineering:** Transforming raw historical race records into high-impact indicators, including custom-designed metrics such as normalized season momentum, recent form adjustments, and driver consistency scores.
- **Exploratory Data Analysis (EDA):** Uncovering and quantifying fundamental race dynamics, including the mathematical correlation of the grid position effect, the statistical impact of multi-stop pit strategies, and historical constructor reliability patterns across the paddock.
- **Predictive Modeling & Evaluation:** Implementing and contrasting an interpretable baseline Linear Regression model against a non-linear, tree-based Random Forest Regressor, followed by a detailed performance comparison using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 metrics.
- **Full-Stack Software Integration:** Deploying the finalized predictive models behind a Flask-based REST API backend that handles request-response validation, which seamlessly feeds an interactive, feature-rich frontend web dashboard complete with timing towers, dynamic impact factor notes, and race-themed user visualizations.

Ultimately, this study serves as a blueprint for how cross-functional development teams can translate complex sports metrics into an integrated, production-ready predictive application.

2 Dataset and Preprocessing

Formula 1 racing results from the 2018-2024 seasons comprise the entire dataset. `finish_position` is the dependent variable in the dataset representing the finishing position of each driver after the race. Training data used for the seasons of 2018-2023 were used in the Linear Regression models; only 2024 is held out as separate test data. Total number of training records in Linear Regression3 was 2456; test records for 2024 had an additional 478 records. Total number of independent

variables in the dataset is four: grid position, recent driver performance, team readiness for the weekend and momentum to the current season. Missing or non-numeric values have all been addressed by changing the type of the variable and filtering using dropna.

3 Feature Engineering

This study utilized feature engineering to transform raw race data into more meaningful variables for the model. Instead of working solely with directly obtained race data, new variables were derived that could better represent driver performance, race week-specific conditions, and the overall trend throughout the season. This resulted in an expanded dataset containing not only basic statistics but also more descriptive indicators that could influence race results.

These newly created variables were specifically designed to reflect drivers' form in recent races, their readiness for a particular race week, and their consistent performance throughout the season. Thus, instead of relying on a single powerful variable like starting position, the model was able to evaluate multiple performance indicators together. This made the prediction process more meaningful and contributed to a clearer interpretation of the factors influencing race finishing position.

Consequently, the feature engineering phase was considered one of the most important steps in the study. The derived variables obtained in this phase supported the predictive power of the created models and provided a stronger analytical structure to the dataset. This resulted in the generation of indicators with higher informational value from raw data and created a more suitable data infrastructure for the modeling process.

Feature	Description	Purpose
driver_form_score	Weighted average of last 3 and last 5 race finishing averages	Captures recent form
weekend_readiness	Combination of qualifying, previous circuit performance, and championship position	Represents race-weekend context
driver_season_momentum	Points before race divided by races before race	Normalizes season performance
driver_recent_improvement	last_5_race_avg_finish - last_3_race_avg_finish	Measures short-term improvement
driver_consistency_score	Absolute difference between last 3 and last 5 race averages	Measures stability

4 Exploratory Data Analysis

The exploratory data analysis (EDA) focused on identifying the key race factors that most strongly influence a driver's final finishing position. By analyzing historical data from 2018 to 2024 (2,979 race entries in total), several critical patterns emerged regarding starting positions, race strategies, and car reliability.

4.1 The Grid Position Effect

A major insight was the strong positive relationship between starting grid position and the final race result. Statistical analysis revealed a Pearson correlation coefficient of 0.696 between the starting grid and the finishing position. Drivers who start closer to the front generally maintain their advantage, while midfield starts create significantly more variation due to traffic and first-lap incidents. Specifically, drivers starting in the top 5 (P1-P5) achieved an average finishing position of 4.33. In contrast, those starting in P6-P10 finished at an average position of 8.54. This means that front-row and top-tier starters finish approximately 4.2 places better on average compared to the upper-midfield. While starting from pole does not guarantee a win, grid position remains the most powerful baseline predictor in our dataset.

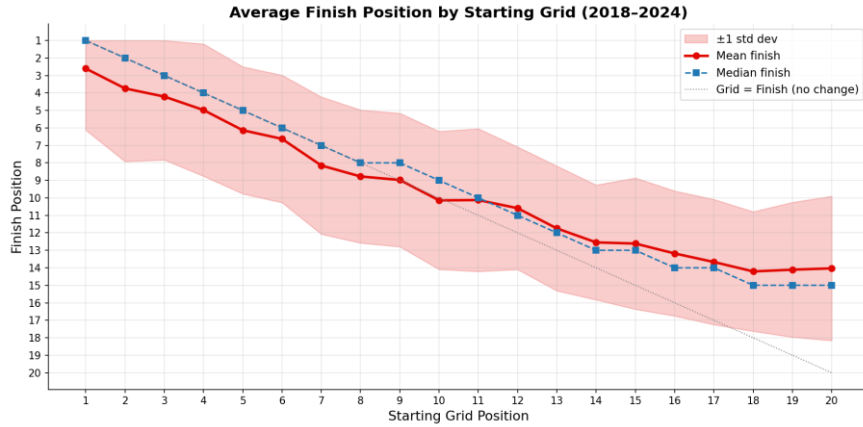


Fig. 1. Average finish position by starting grid position.

4.2 Pit Stop Strategy Insights

Tire degradation and pit stop timing are vital tactical components. Analyzing modern F1 seasons (2018–2024), we found that a one-stop strategy is generally the optimal and most common standard.

1. Stop Strategy: Average finish of 8.85 (1,082 occurrences).
2. Stop Strategy: Average finish of 9.44 (1,008 occurrences).
3. Stop Strategy: Average finish of 10.01 (330 occurrences).

4. Stop Strategy: Average finish of 10.34 (96 occurrences).

The data shows a +0.60 position gap between two-stop and one-stop strategies, indicating that drivers who pit twice or more tend to either start further back or are forced into sub-optimal strategies due to unexpected tire degradation, damage, or changing weather.

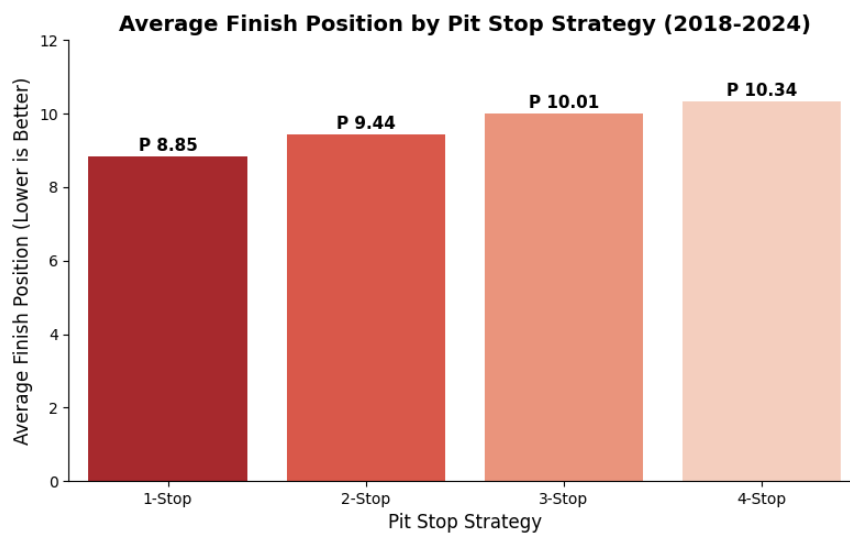


Fig. 2. Impact of pit stop strategy on average finish position.

4.3 Constructor Reliability and DNF Rates

To win in Formula 1, you first have to finish. Constructor reliability is a decisive championship factor. We calculated the 'Did Not Finish' (DNF) rates for teams with over 30 starts in our dataset. The findings highlight a clear divide between top-tier teams and the rest of the pack:

- **Mercedes:** 6.7% DNF rate (20 DNFs in 298 starts).
- **McLaren:** 8.7% DNF rate (26 DNFs in 298 starts).
- **Ferrari:** 9.4% DNF rate (28 DNFs in 298 starts).
- **Red Bull:** 10.1% DNF rate (30 DNFs in 298 starts).

Conversely, teams struggling at the back of the grid suffered from much poorer reliability. For instance, Haas F1 Team had a 14.8% DNF rate, Williams recorded

17.8%, and Toro Rosso faced a peak 19.0% DNF rate. Top constructors consistently keep both cars running to the flag, which heavily skews finishing predictions in favor of drivers in reliable machinery.

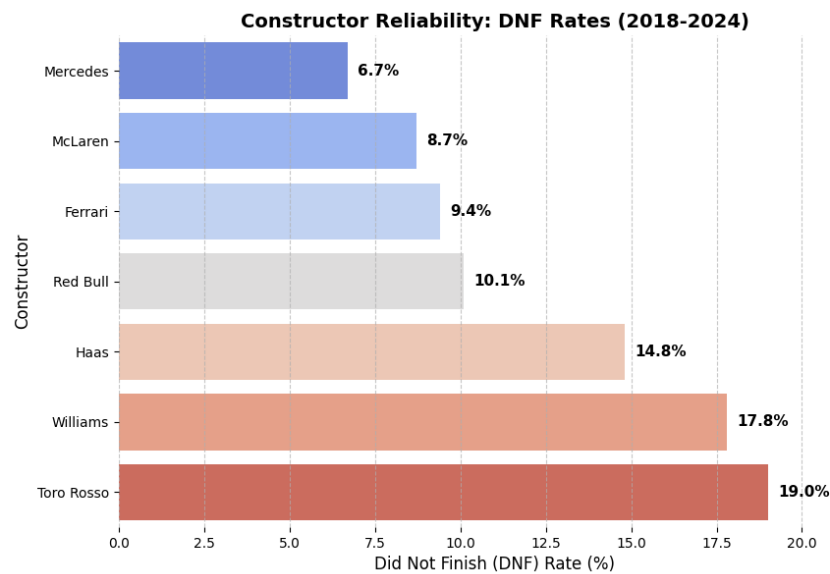


Fig. 3. Did Not Finish (DNF) percentage comparison across major constructors.

4.4 Engineered Features Highlight: Season Momentum

To capture a driver's current form without bias toward late-season races, we engineered a new feature: `driver_season_momentum`. This calculates the average championship points earned per race *before* a specific event. By dividing the raw season points by the number of races run, we created a comparable momentum rate across the entire calendar, preventing data leakage and providing the models with a normalized performance metric.

5 Predictive Modeling

5.1 Linear Regression

The Linear Regression model was chosen as the basic prediction model in this study and used as a starting point for comparison with other models. The most important reason for choosing this method is its simple and interpretable structure. Linear Regression models the relationship between independent variables and the target variable through a linear structure, and as a result, it numerically shows the

effect of each variable on the outcome. This provides a significant benefit not only for generating predictions but also for interpreting the factors affecting the race result in a more understandable way. Especially in studies where results dependent on numerous factors, such as sports data, the explainability of the model strengthens the evaluation process.

In the model we used, the `finish_position`, which represents the order in which the drivers finished the race, was used as the target variable. In the prediction process, three basic variables that we believe determine the race result were included in the model. The first of these is the `grid` variable. The `grid` shows the position in which the driver started the race, and since the effect of starting position on the result is quite high in Formula 1 races, it is an important input for the model. The second variable, `driver_form_score`, was created to represent the average performance of the driver in recent races. Thanks to this variable, we arrived at a more balanced result based not only on a single race outcome but also on the driver's recent form. The third variable, `weekend_readiness`, was used to represent how ready the driver was for the relevant race weekend. This variable was created by evaluating different indicators together, such as qualifying performance, the driver's previous performance at the same track, and their position in the championship standings.

Another reason for using the Linear Regression model in this study is that it allows for an understanding of the fundamental trends in the data before moving on to more complex models. Although the model offers a basic prediction approach, it contributes to explaining the relationship between race outcome and grid position, recent form, and weekend readiness to a certain extent. Thus, both the effect of important variables in the dataset can be observed, and a basis for comparison is created for more advanced models to be applied later. In this respect, Linear Regression was evaluated in the study not only as a technical model but also as an analytical tool that facilitates the interpretation of the results.

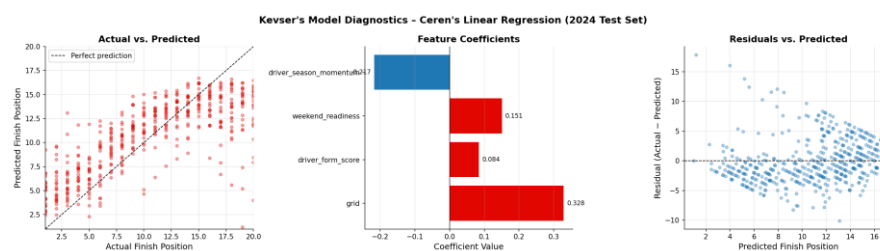


Fig. 4. Linear Regression diagnostics for the 2024 test set.

Looking at the tables in Fig. 4, we can see the diagnostic results of the Linear Regression model on the 2024 test data. The first graph compares the actual

finishing positions with the finishing positions predicted by the model. The clustering of points around the reference line shows that the model was able to largely capture the basic trend in race results. It is observed that the predictions are closer to the actual values, especially in results in the middle ranks. As a result, deviations increase in race scenarios with very good or very bad results. This indicates that the Linear Regression model offers sufficient performance in explaining normal race dynamics, but is more limited in predicting unusual race results.

5.2 Random Forest Regressor

The main purpose of using Random Forest is to find relationships between race indicators and finishing position. In addition, the driver_recent_improvement and driver_consistency_score properties have been added to this model. In order to support the interpretation of the model, the feature importance analysis was used.

Random Forest implementation details. The random forest method was applied as the tree-based model for the project. This model uses the starting position, driver form, race week preparation, season momentum and predicts the finishing position using with two additional trend-based features. This model is useful because the factors affecting the outcomes of Formula 1 races are non-linear. For example, the driver's starting position effect may vary depending on the driver's recent performance or momentum.

Table RF-1. Random Forest model configuration.

Parameter	Value
Estimator	RandomForestRegressor
Training seasons	2018-2023
Test season	2024
Target variable	finish_position
Input features	grid, driver_form_score, weekend_readiness, driver_season_momentum, driver_recent_improvement, driver_consistency_score
Number of trees	200
Maximum depth	8
Random state	42
Prediction range	Clipped to P1-P20

Additional Random Forest features

Feature	Formula	Interpretation
driver_recent_improvement	$\text{last_5_race_avg_finish} - \text{last_3_race_avg_finish}$	Captures whether the driver's latest three-race form is improving compared with the broader five-race window.
driver_consistency_score	$\text{abs}(\text{last_3_race_avg_finish} - \text{last_5_race_avg_finish})$	Measures recent volatility; lower values indicate more stable finishing performance.

driver_recent_improvement compares the average finishing positions of the last five races and the last three races. In Formula 1, a lower finishing position indicates a better result. This feature indicates whether a driver's recent performance on the model is on the rise or on the decline.

driver_consistency_score finds the difference between short-term and medium-term completion performance and measures the absolute value of this difference. Achieving a lower value indicates that the driver's performance has been consistent and stable in recent periods. However, obtaining a higher value indicates fluctuations in driver performance. Therefore, predicting the final finishing position becomes difficult.

Random Forest metric interpretation

Table RF-2. Random Forest performance on the 2024 test season.

Metric	Value	Interpretation
MAE	3.033	The model is wrong by about three finishing positions on average.
RMSE	3.881	Large race-outcome mistakes are penalized more strongly.
R2	0.546	The model explains a moderate amount of finish-position variation on the 2024 test set.

The fact that the MAE value is 3.033 means that the Random Forest model deviates from the actual results by an average of 3 finishing positions when

predicting the 2024 race results.This is an understandable result for Formula 1.Because the final rankings can change very quickly for many reasons.

The RMSE value being 3.881 indicates that it is higher than the MAE value because RMSE gives more weight to larger errors. This situation indicates that the predictions are largely reasonably close to the actual results. However, it is stated that some race results lead to larger prediction errors. Situations such as the driver starting from the front rows and then finishing the race much further behind than expected increase the model's margin of error.

The R² score being 0.546 indicates that the model has a moderate level of explanatory power. The model is able to capture important patterns, especially from pre-race features such as starting position (grid position) and season momentum (season momentum).However, since it cannot fully predict unexpected events that occur on race day, it cannot fully explain the variability in results.

Feature importance interpretation

Table RF-3. Random Forest feature importance ranking.

Rank	Feature	Importance
1	grid	0.478
2	driver_season_momentum	0.241
3	weekend_readiness	0.102
4	driver_form_score	0.085
5	driver_recent_improvement	0.051
6	driver_consistency_score	0.043

The feature importance results show that the grid position variable is the most dominant predictor, with an importance value of 0.478.This result is consistent with the Formula 1 field knowledge. Because starting from the front rows reduces the likelihood of getting stuck in traffic and exposure to in-race incidents.

The second most important feature was driver_season_momentum with an importance value of 0.241.This situation demonstrates that the driver's overall season-long points-scoring performance is also a strong indicator in predicting race results.

The following property materiality graph also visualizes the same pattern: grid position provides the highest contribution, followed by driver_season_momentum. Features such as driver_recent_improvement and driver_consistency_score, although having lower importance values, provide useful trend information to the model.

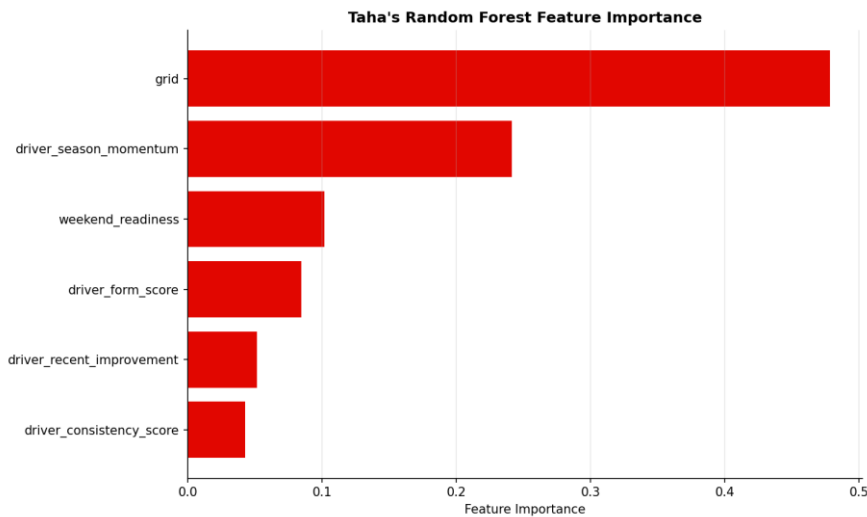


Fig. 6. Random Forest feature importance.

6 Model Evaluation

6.1 Evaluation Methodology

To measure how well each model learned from the data, an evaluation module (`evaluation.py`) was developed to calculate three common regression metrics using NumPy: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Both models were then tested using data from the 2024 Formula 1 season, which contained 478 race entries that were not included in the training process.

Each metric measures performance in a different way. MAE shows the average difference between the predicted and actual finishing positions, making it easy to understand prediction accuracy. RMSE also measures prediction error but gives greater weight to larger mistakes, meaning predictions that are far from the actual result have a stronger effect on the score. R^2 measures how much of the variation in race results can be explained by the model. A value closer to 1 indicates that the model is better at explaining and predicting finishing positions.

6.2 Results

Both models were evaluated during the 2024 season. Lower MAE and RMSE mean more accurate predictions, while a higher R^2 means the model captures more of the real pattern in the data.

Model	MAE	RMSE	R2
Linear Regression	2.897	3.753	0.576
Random Forest	3.033	3.881	0.546

Linear Regression performed better than Random Forest on all three metrics. Its MAE was about 2.9, which means the model's predictions were usually less than three positions away from the actual finishing position. This is a good result because Formula 1 races can be very unpredictable. The model's R^2 score was 0.576, meaning it could explain about 57.6% of the differences in the final race results using the features included in the model.

6.3 Why Linear Regression Outperformed Random Forest

At first glance, these results may seem surprising because Random Forest is often considered the more powerful algorithm. However, the outcome becomes clearer when the characteristics of the data are examined.

Linear Regression used four features: grid position, driver form score, weekend readiness, and season momentum. Analysis of the model coefficients showed that grid position (0.328) and season momentum (-0.217) had the strongest influence on the predictions. These features appeared to have a largely linear relationship with finishing position, which suited the Linear Regression model well.

Random Forest used the same four features along with two additional features: recent improvement and consistency score. However, the extra complexity did not improve performance. The model may have slightly overfitted patterns from previous seasons that did not remain consistent during the 2024 season. In this case, the simpler model was better able to capture the underlying relationships in the data.

6.4 Model Comparison Charts

After both models were evaluated, their results were collected and used to automatically generate three bar charts showing RMSE, MAE, and R^2 . The evaluation metrics were also saved in a file named `model_comparison_metrics.csv`, allowing the results to be reviewed easily at a later stage.

Figures 7, 8, and 9 present a side-by-side comparison of the two models. All three charts show the same pattern: Linear Regression performed slightly better than Random Forest across every evaluation metric. Although the differences were relatively small, Linear Regression consistently achieved stronger results.

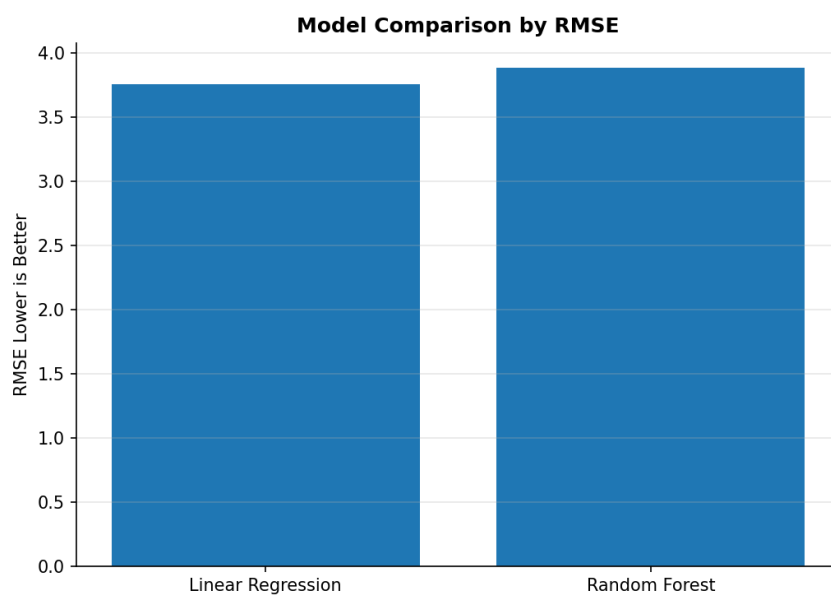


Fig. 7. RMSE comparison between Linear Regression and Random Forest.

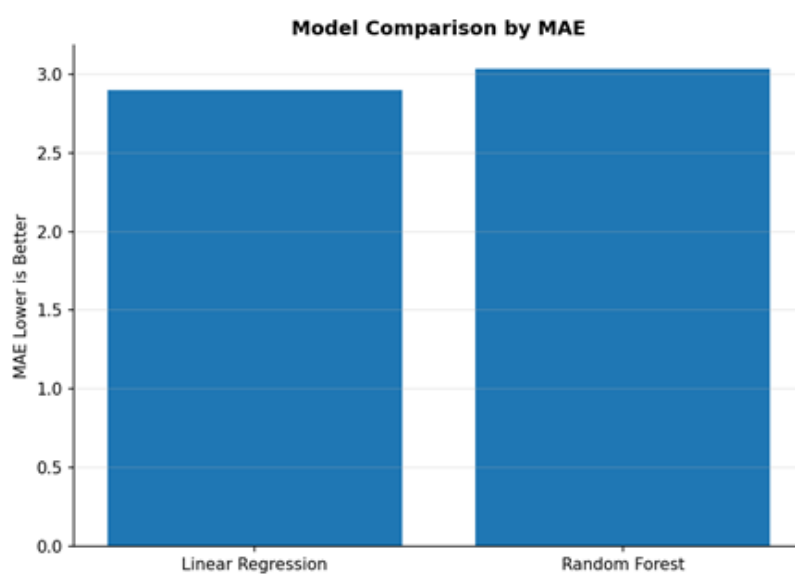


Fig. 8. MAE comparison between Linear Regression and Random Forest

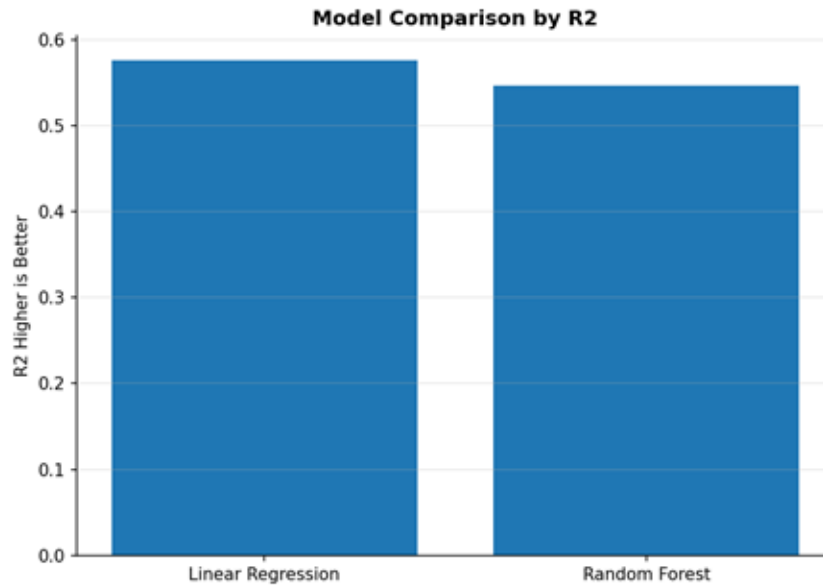


Fig. 9. R² comparison between Linear Regression and Random Forest

Error Distribution Analysis.

Figure 10 shows the error distribution of the Random Forest model. The histogram highlights an important pattern: the errors are not evenly spread around zero. Instead, the distribution is slightly shifted to the left, which means the model tends to overestimate finishing positions more often than it underestimates them. In other words, the model frequently predicts that a driver will finish in a better position than they actually do.

Most errors fall between -5 and 0, while positive errors occur less often but can occasionally be much larger, reaching up to +15 positions. This type of bias is not unusual in Formula 1 prediction models. The model was trained on previous seasons where driver performance was generally more stable, whereas the 2024 season contained more unexpected outcomes and variation.

Ideally, prediction errors should be evenly distributed around zero. Since the distribution is slightly skewed, the results suggest that a small prediction bias remains and could be addressed in future improvements of the model.

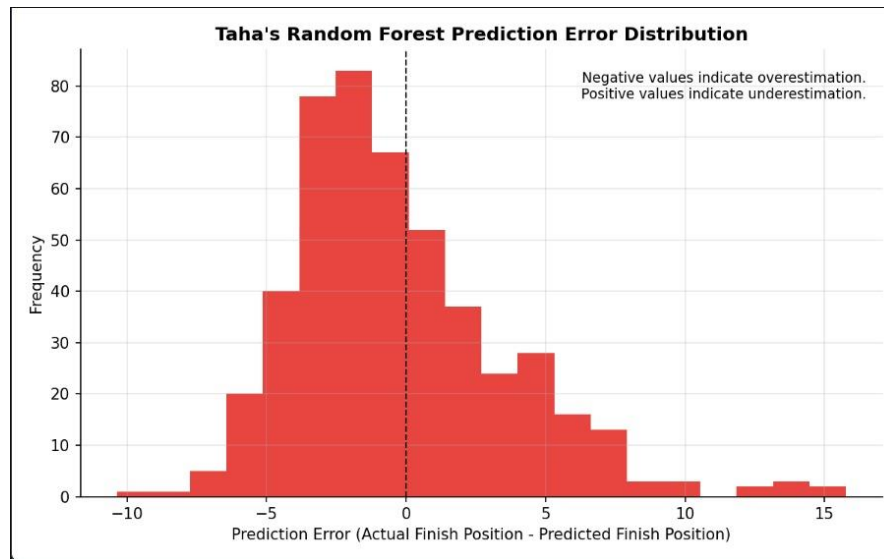


Fig. 10. Random Forest prediction error distribution

6.5 API Integration Testing

The final stage of the project involved verifying that the trained model functioned correctly within the Flask API environment. A test suite (`test_api.py`) was developed containing seven test cases that covered the main API functions. These tests included checking the root and `/health` endpoints, submitting a complete prediction request with all features, submitting a minimal request containing only grid position, and verifying that invalid inputs were correctly rejected with a 400-error response.

A multi-driver simulation was also performed using Verstappen, Hamilton, and Alonso at different grid positions to ensure that predictions remained within the realistic range of 1 to 20 finishing positions. All seven test cases were completed successfully, confirming that the API and prediction system operated as expected.

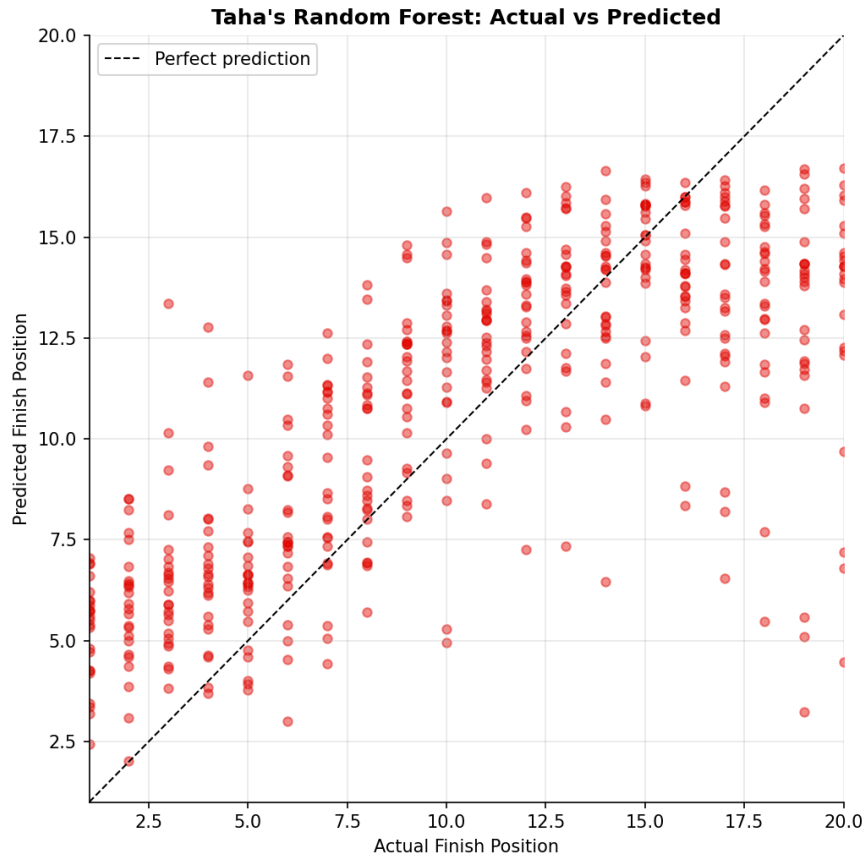


Fig. 11. Random Forest actual vs. predicted finish positions.

7 Backend API and Frontend Dashboard

A Flask backend API was developed to connect the trained prediction workflow to the frontend interface. The API provides a bridge between the machine learning model and the user-facing dashboard. It exposes GET /health for checking whether the service is running correctly and POST /predict for generating a finishing position prediction from user-provided race indicators. When a request is sent to the /predict endpoint, the backend processes the input values, applies the required engineered feature logic, and returns the predicted finishing position in JSON format.

The frontend dashboard sends user inputs such as starting grid position, driver form score, weekend readiness, and season momentum to this endpoint. After receiving the API response, it displays the predicted result in a clear and

interactive format. The dashboard includes input fields, an animated prediction result, a timing tower, factor notes, model report charts, F1-themed styling, race lights animation, and car sound effects. These elements were added to make the prediction output easier to understand and more engaging for users.

Overall, this part of the system represents the integration layer of the project. The backend handles model-based prediction, while the frontend transforms the result into a visual and user-friendly experience. This connection allows the project to move beyond static analysis files and become an interactive application that users can test with different race scenarios.

8 Discussion

This project successfully demonstrates how raw, historical sports data can be transformed into a usable predictive software product. The modeling pipeline seamlessly produces predictions and evaluation reports, while the Flask API and frontend dashboard make these insights accessible through a highly interactive, user-facing interface.

Our Exploratory Data Analysis confirmed that Formula 1 is deeply hierarchical. Predictors like the starting grid, constructor reliability (with teams like Mercedes showing remarkably low DNF rates of 6.7%), and seasonal momentum dictate the baseline outcomes. The performance of our Linear Regression model—achieving an MAE of 2.897—suggests that for front-runners, the relationship between these predictors and the final result is strongly linear. Top drivers in reliable cars who qualify well usually finish well.

However, the main limitation of our study lies in the midfield. Both models struggled to accurately predict the finish positions for drivers starting between P6 and P15. As our pit-stop analysis indicated, mid-race chaos—such as forced two-stop strategies, traffic, safety cars, mechanical failures, and unpredictable tire degradation—injects heavy variance into the results. A driver can lose 5 to 10 positions in seconds due to factors that are extremely difficult to represent numerically before the race begins.

Future Work: To bridge the gap in predicting midfield chaos, future models could incorporate real-time race data. Adding features for specific tire compounds (Soft/Medium/Hard start), exact pit stop timings, dynamic weather conditions, track temperature, and circuit-specific characteristics (e.g., street circuits vs. high-speed tracks) would allow for non-linear models to perform much better. Furthermore, utilizing deep learning architectures to process historical telemetry data could add an entirely new layer of accuracy to F1 race prediction.

9 Conclusion

Finally, the project combines Formula 1 data analysis, predictive modeling, backend API development, and frontend visualization into a single system. Throughout the project, historical race data was processed, transformed into meaningful features, and used to train predictive models to predict race finishing positions. In this case, the project did not limit itself to a theoretical machine learning application, but transformed the analysis into a practical software solution with which users can directly interact.

The system we developed predicts race finishing positions using selected race weekend indicators such as grid position, driver form, weekend preparation, and season-related performance signals. These model outputs are then visualized with an interactive dashboard supported by a backend API and a user-facing frontend interface.

In conclusion, our study has shown that race results in Formula 1 can be meaningfully modeled using data-driven methods, despite the presence of many unpredictable factors such as weather conditions, safety car periods, technical failures, and strategy changes. Despite these limitations, the proposed system offers a useful and interpretable approach to race result prediction. Therefore, the project has served as a successful example of combining machine learning, software integration, and domain-specific sports analytics in a single application.

10 Run Instructions

1. Install the required Python dependencies from requirements.txt.
2. Run the backend API with `python3 src/api.py`.
3. Open `frontend/index.html` with Live Server or a local static server.
4. Enter race-weekend values and click Predict finish.
5. Review the generated prediction, timing tower, impact notes, and report charts.

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